

Total Marks: 30	Total Questions: 17	Due: 16 Jul 2026
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Name: _____ Roll Number: _____ Section: _____

General Instructions:

1. Time Limit Only 1 Hour

SECTION A	[10 marks]
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Choose the most appropriate answer from the given options.

- Q1.

What is the speed of light in a vacuum?

A. 2×10^8 m/s

B. 3×10^8 m/s

C. 2.5×10^8 m/s

D. 3.5×10^8 m/s

Easy

[1 mark]
- Q2.

What is the phenomenon of light bending called?

A. Reflection

B. Refraction

C. Diffraction

D. Dispersion

Medium

[1 mark]
- Q3.

What type of lens converges light rays?

A. Convex lens

B. Concave lens

C. Plane mirror

D. Prism

Easy

[1 mark]
- Q4.

What is the color of light with the shortest wavelength?

A. Red

B. Orange

C. Yellow

D. Violet

Medium

[1 mark]
- Q5.

What is the term for the ability of a material to transmit light?

A. Opacity

B. Transparency

C. Reflectivity

D. Refractivity

Hard

[1 mark]
- Q6.

What is the process of light passing through a narrow opening called?

A. Diffraction

B. Refraction

C. Reflection

D. Dispersion

Medium

[1 mark]
- Q7.

What type of mirror always produces a virtual image?

A. Plane mirror

B. Concave mirror

C. Convex mirror

D. Spherical mirror

Easy

[1 mark]
- Q8.

What is the term for the spreading of light as it passes through a narrow opening?

A. Diffraction

B. Refraction

C. Dispersion

D. Scattering

Hard

[1 mark]
- Q9.

What is the color of light with the longest wavelength?

A. Red

B. Orange

C. Yellow

D. Violet

Medium

[1 mark]

Q10. What is the term for the bending of light as it passes from one medium to another?

Easy

[1 mark]

- A. Reflection
- C. Diffraction

- B. Refraction
- D. Dispersion

SECTION B

[8 marks]

Answer all questions briefly. Each question carries 2 marks.

Q11. What is the difference between a convex lens and a concave lens?

Easy

[2 marks]

Answer: _____

Q12. What is the principle of total internal reflection?

Medium

[2 marks]

Answer: _____

Q13. What is the term for the ability of a material to absorb light?

Hard

[2 marks]

Answer: _____

Q14. What is the difference between a real image and a virtual image?

Medium

[2 marks]

Answer: _____

SECTION C

[12 marks]

Answer all questions briefly. Each question carries 4 marks.

Q15. What is the process of light passing through a prism and what are the resulting colors?

Hard

[4 marks]

Answer: _____

Q16. What is the term for the ability of a material to transmit light and what are the factors that affect it?

Medium

[4 marks]

Answer: _____

Q17. What is the difference between a concave mirror and a convex mirror and what are their uses?

Easy

[4 marks]

Answer: _____